

Closed-form update for the KL-regularised step

We wish to maximise, for a fixed $\beta > 0$,

$$J_k(\pi) = \mathbb{E}_{q \sim Q} \mathbb{E}_{y \sim \pi(\cdot|q)} [f_k(q, y)] - \beta \text{KL}(\pi \| \pi_k), \quad (1)$$

where the reward is the first variation of the entropy,

$$f_k(q, y) = -(1 + \log \pi_k(y|q)).$$

Because both terms in (1) are conditional on the query q , we can solve, for every fixed q ,

$$\begin{aligned} \max_{\pi(\cdot|q)} \quad & \sum_y \pi(y|q) f_k(q, y) - \beta \sum_y \pi(y|q) \log \frac{\pi(y|q)}{\pi_k(y|q)} \\ \text{s. t.} \quad & \sum_y \pi(y|q) = 1, \quad \pi(y|q) \geq 0. \end{aligned}$$

Plugging the reward into the objective above,

$$L(\pi) = \sum_y \pi(y|q) (\beta - 1) \log \pi_k(y|q) - \beta \sum_y \pi(y|q) \log \pi(y|q).$$

Introduce the multiplier $\lambda(q)$:

$$\Lambda(\pi, \lambda) = L(\pi) + \lambda(q) \left(\sum_y \pi(y|q) - 1 \right)$$

Setting $\partial \Lambda / \partial \pi(y|q) = 0$ gives

$$(\beta - 1) \log \pi_k(y|q) - \beta (\log \pi(y|q) + 1) + \lambda(q) = 0, \quad \forall y.$$

Solve for $\pi(y|q)$:

$$\log \pi(y|q) = \frac{\lambda(q) - \beta}{\beta} + \left(1 - \frac{1}{\beta}\right) \log \pi_k(y|q)$$

Exponentiating,

$$\pi(y|q) = \alpha(q) \pi_k(y|q)^s, \quad s := 1 - \frac{1}{\beta}, \quad (1)$$

with $\alpha(q) = \exp((\lambda(q) - \beta)/\beta)$

Impose $\sum_y \pi(y|q) = 1$ to normalise:

$$\alpha(q) = \left(\sum_y \pi_k(y|q)^s \right)^{-1}.$$

Combining, we get the explicit update

$$\pi_{k+1}(y|q) = \frac{\pi_k(y|q)^{1-\frac{1}{\beta}}}{\sum_{y'} \pi_k(y'|q)^{1-\frac{1}{\beta}}}$$

Note that for $\beta > 1 \Rightarrow 0 < s < 1$ each iteration flattens the distribution, increasing entropy until it approaches uniform.